

Solutions to Worksheet #4

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1. **Solution.** Recall that

$$X_G = \sum_{\kappa} \prod_{v \in V(G)} x_{\kappa(v)},$$

where the sum is over proper colorings by positive integers. Let σ be any permutation of the positive integers. Replacing every color i by $\sigma(i)$ is a bijection from proper colorings to proper colorings. It permutes the variables in every monomial but leaves the whole sum unchanged. Thus X_G is invariant under every permutation of the variables, so it is symmetric.

2. **Solution.** Let e be any edge of C_4 . Deleting e produces the path P_4 , which is a tree on four vertices, while contracting e produces the triangle C_3 . Therefore

$$\chi_{C_4}(n) = n(n-1)^3 - n(n-1)(n-2).$$

Simplifying,

$$\chi_{C_4}(n) = n(n-1)(n^2 - 3n + 3) = n^4 - 4n^3 + 6n^2 - 3n.$$

3. **Solution.** We use Stanley's power-sum formula

$$X_G = \sum_{A \subseteq E(G)} (-1)^{|A|} p_{\lambda(A)},$$

where $\lambda(A)$ is the partition of the component sizes of the spanning subgraph $(V(G), A)$.

- (a) Deleting the chosen edge gives P_4 . Among its three edges, there are three one-edge subsets, two adjacent two-edge subsets, one disjoint two-edge subset, and one three-edge subset. Hence

$$X_{P_4} = p_{1,1,1,1} - 3p_{2,1,1} + 2p_{3,1} + p_{2,2} - p_4.$$

- (b) Contracting the edge gives $C_3 = K_3$. The subset formula gives

$$X_{C_3} = p_{1,1,1} - 3p_{2,1} + 2p_3.$$

- (c) The two terms in a deletion-contraction formula do not even have the same degree: X_{P_4} has degree 4, while X_{C_3} has degree 3. More explicitly,

$$X_{C_4} = p_{1,1,1,1} - 4p_{2,1,1} + 4p_{3,1} + 2p_{2,2} - 3p_4,$$

which is not $X_{P_4} - X_{C_3}$. Thus the ordinary deletion-contraction relation does not hold for the chromatic symmetric function.

4. Solution.

- (a) By the binomial theorem,

$$\begin{aligned} n(n-1)^{k-1} &= n \sum_{i=0}^{k-1} \binom{k-1}{i} n^i (-1)^{k-1-i} \\ &= \boxed{\sum_{j=1}^k (-1)^{k-j} \binom{k-1}{j-1} n^j}. \end{aligned}$$

- (b) Taking $j = k - 1$, the coefficient is

$$\boxed{-\binom{k-1}{k-2} = -(k-1)}.$$

- (c) There is a missing minus sign in the worksheet statement. For a simple graph,

$$\boxed{[n^{k-1}] \chi_G(n) = -|E(G)|}.$$

We prove this by induction on the number of edges. For the graph with no edges, $\chi_G(n) = n^k$, and both sides are 0. Choose an edge e . By induction, the coefficient of n^{k-1} in $\chi_{G-e}(n)$ is $-(|E(G)| - 1)$. The contracted graph has $k - 1$ vertices, so its chromatic polynomial is monic of degree $k - 1$. Consequently, deletion-contraction gives

$$[n^{k-1}] \chi_G(n) = -(|E(G)| - 1) - 1 = -|E(G)|.$$

- (d) If $\chi_G(n) = n(n-1)^{k-1}$, part (b) says that the coefficient of n^{k-1} is $-(k-1)$. By part (c), G has $k-1$ edges. Since G is connected, a connected graph on k vertices with $k-1$ edges is a tree.

5. **Solution.** Call the graphs G_1 and G_2 in their displayed order. In either graph there are four independent pairs. Thus a coloring of type $(2, 1, 1, 1)$ is obtained by choosing one of those pairs and then assigning the three singleton colors in $3!$ ways, giving coefficient $4 \cdot 3! = 24$.

There are two ways in either graph to choose two disjoint independent pairs. Since the two pair-colors can be interchanged, the coefficient of

$m_{2,2,1}$ is $2 \cdot 2! = 4$. No independent set has size 3, and a coloring with all colors distinct contributes $5! = 120$. Therefore the two graphs have the same answer:

$$X_{G_1} = X_{G_2} = 4m_{2,2,1} + 24m_{2,1,1,1} + 120m_{1,1,1,1,1}.$$

6. **Solution.** In the power expansion, the coefficient of p_n is

$$\sum_{\substack{A \subseteq E(G) \\ (V(G), A) \text{ connected}}} (-1)^{|A|}.$$

It is zero when G is disconnected, because no edge subset can connect all the vertices. When G is connected, the signed sum is nonzero; up to the sign $(-1)^{n-1}$, it counts acyclic orientations having a unique sink at any fixed vertex. At least one such orientation is obtained by ordering vertices by their distance from the chosen sink and directing edges toward smaller distance. Consequently,

$$G \text{ is connected if and only if } [p_n]X_G \neq 0.$$

More generally, the smallest number of parts among partitions indexing nonzero power-sum terms is the number of connected components.

7. **Solution.** The term $p_{2,1^{n-2}}$ in the subset expansion can arise only from a subset containing exactly one edge. Every edge contributes $-p_{2,1^{n-2}}$. Hence

$$[p_{2,1^{n-2}}]X_G = -|E(G)|.$$

If $X_G = X_H$, these coefficients agree, so G and H have the same number of edges.

8. **Solution.** In a proper coloring of K_n , all vertices receive distinct colors. For each set of n colors there are $n!$ assignments to the vertices, so

$$X_{K_n} = n!e_n.$$

In particular,

$$X_{K_1} = e_1, \quad X_{K_2} = 2e_2, \quad X_{K_3} = 6e_3, \quad X_{K_4} = 24e_4, \quad X_{K_5} = 120e_5.$$

9. **Solution.** Let $e = uv$, where u is the leaf and v is its neighbor.

- (a) Deleting e leaves u isolated. Thus $G \setminus e = (G - u) \sqcup K_1$.
- (b) Dot-contraction contracts u and v and then adds an isolated vertex. Contracting a leaf edge simply removes the leaf, so the result is again $(G - u) \sqcup K_1$.
- (c) The graphs in (a) and (b) are isomorphic.

- (d) Leaf-contraction contracts u and v and then attaches a new leaf to the merged vertex. This recreates a graph isomorphic to G .
- (e) The deletion-near-contraction relation is

$$X_G = X_{G \setminus e} - X_{\text{dot}(G,e)} + X_{\text{leaf}(G,e)}.$$

For a leaf edge, the first two terms cancel and the last term is X_G , so the relation reduces to the tautology $X_G = X_G$. It makes no progress. This is why the recursive algorithm uses internal edges.

10. **Solution.** Write $\text{st}_r = X_{St_r}$ and $\text{st}_\lambda = \prod_i \text{st}_{\lambda_i}$. There are three trees on five vertices: St_5 , the tree T_5 with degree sequence $(3, 2, 1, 1, 1)$, and the path P_5 .

For T_5 , apply DNC to the unique internal edge adjacent to the degree-3 vertex. The deletion, dot-contraction, and leaf-contraction terms are respectively star forests of types $(3, 2)$, $(4, 1)$, and (5) . Thus

$$X_{T_5} = \text{st}_5 - \text{st}_{4,1} + \text{st}_{3,2}.$$

Similarly, applying DNC to the middle edge of P_4 gives

$$X_{P_4} = \text{st}_{2,2} - \text{st}_{3,1} + \text{st}_4.$$

Applying DNC to an internal edge of P_5 then gives

$$\begin{aligned} X_{P_5} &= \text{st}_{3,2} - X_{P_4} \text{st}_1 + X_{T_5} \\ &= \text{st}_5 - 2\text{st}_{4,1} + 2\text{st}_{3,2} + \text{st}_{3,1,1} - \text{st}_{2,2,1}. \end{aligned}$$

Finally, by definition,

$$X_{St_5} = \text{st}_5.$$

11. **Solution.** Call the displayed graphs H_1 and H_2 . The subset formula gives the same power-sum expansion for both:

$$\begin{aligned} X_{H_i} &= p_1^6 - 6p_{2,1^4} + 6p_{2,2,1,1} - p_{2,2,2} + 8p_{3,1,1,1} \\ &\quad - 6p_{3,2,1} - 9p_{4,1,1} + 3p_{4,2} + 6p_{5,1} - 2p_6. \end{aligned}$$

To convert to the star basis, one may use

$$p_n = \sum_{r=0}^{n-1} (-1)^r \binom{n-1}{r} \text{st}_{r+1, 1^{n-1-r}}$$

and multiply this identity for the parts of each power-sum index. After collecting terms,

$$\begin{aligned} X_{H_1} = X_{H_2} &= 2\text{st}_6 - 4\text{st}_{5,1} + 3\text{st}_{4,2} + 2\text{st}_{4,1,1} \\ &\quad - 3\text{st}_{3,2,1} + \text{st}_{2,2,2}. \end{aligned}$$

Thus these two nonisomorphic graphs have the same chromatic symmetric function.